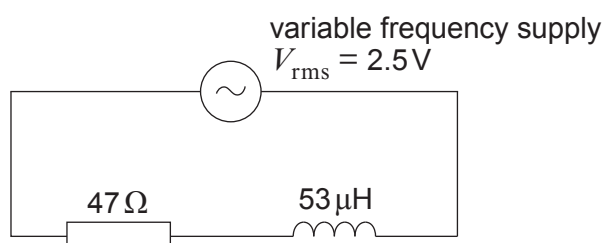


Option A – Alternating Currents

7. (a) (i) Explain why the pd across the resistor decreases as the frequency of the supply increases. [2]

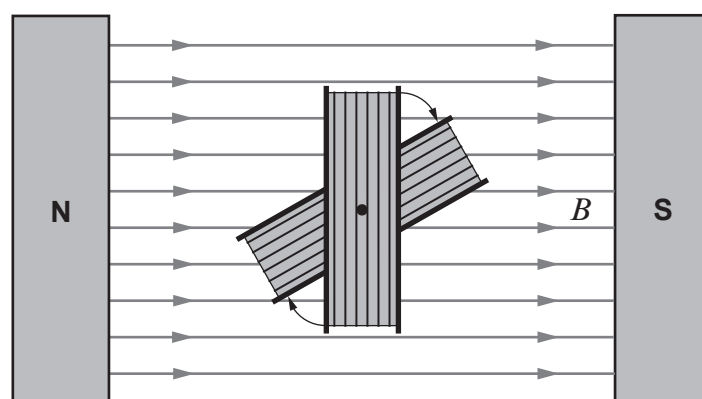


- (ii) Calculate the frequency when the pd across the resistor and inductor are equal. [2]

- (iii) Calculate the rms pd across the resistor when it is equal to the rms pd across the inductor. [2]



- (b) A square coil of N turns and cross-sectional area, A , is rotated with angular velocity, ω , in a magnetic field of flux density, B .



The peak induced emf is given by the equation:

$$V = \omega BAN$$

- (i) Explain in terms of Faraday's law why the emf is proportional to each of B and A . [3]

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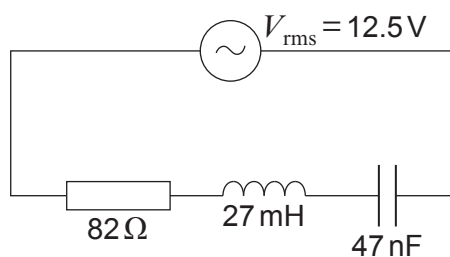
- (ii) Explain at which point in the cycle the peak emf occurs. [1]

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- (c) Calculate the rms current in the following circuit:



- (i) at resonance;

[1]

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- (ii) at twice the resonance frequency.

[4]

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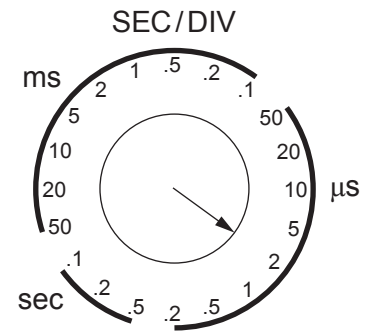
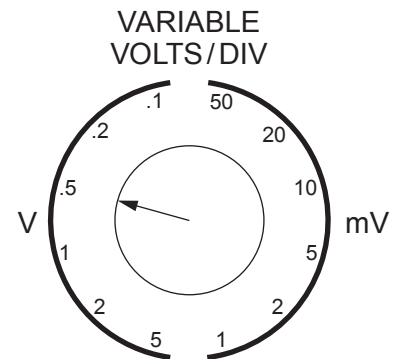
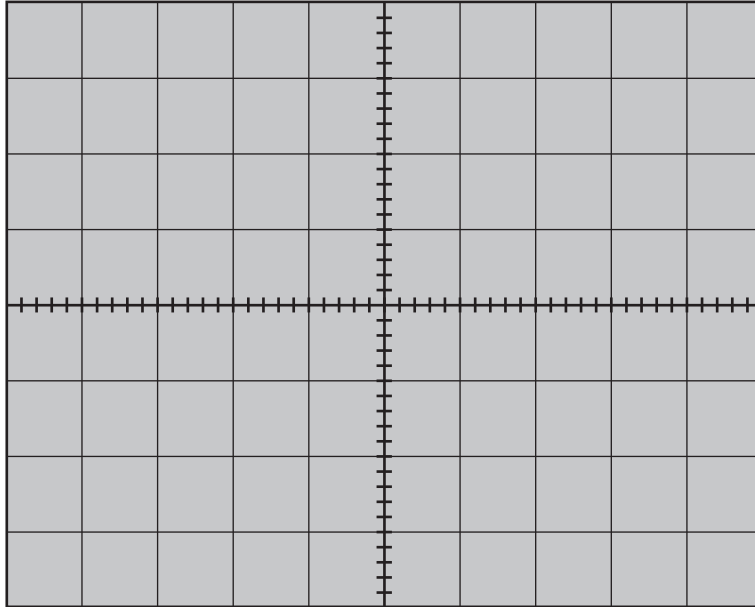
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Examiner
only

- (d) Tomos wants to display a wave trace on an oscilloscope for an a.c. power supply of frequency 67.5 kHz and rms pd 240 mV. Determine whether or not the settings of the oscilloscope shown below are appropriate.

[5]



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